

Who Earns Over \$50K?

Predicting Income from US Census Data

A Deep-Dive Into 347 Million Americans · 1994–1995 Census Bureau Data

199K+

Training records

40

Features

347M

Americans represented

3

Models compared

Today's Agenda

01

Business Context

Why predict income? The allocation problem.

02

The Dataset

347M Americans, 40 signals, two years of census data.

03

Key Findings from EDA

Education, age, capital income, and demographic patterns.

04

Modelling Approach

Three competing models: Logistic Regression, Random Forest, LightGBM.

05

Results & Interpretation

ROC-AUC 0.955 · PR-AUC 0.677 · SHAP explainability.

06

Takeaways & Next Steps

Findings, honest limitations, and deployment considerations.

Why Predict Income? The Allocation Problem

\$4T+

Federal funding allocated annually using Census income data — hospitals, schools, fire departments, infrastructure

Policy

A predictive model flags under-served populations and identifies data quality issues at scale

Question

What characteristics predict whether a person earns $\geq$ \$50,000/year?

Dual Role

Both a classification problem and a policy analysis tool — technical rigor meets real-world impact

The Data: 347 Million Americans in Two Years

199,523

Training records

99,762

Test records

40

Features per individual

347M

Americans represented

Feature Categories

Continuous

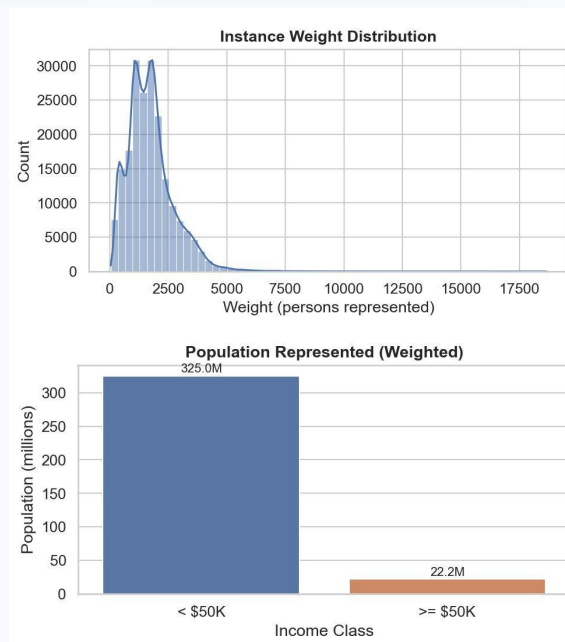
Age, weeks worked, wage/hr, capital gains, capital losses, dividends

Categorical

Education, marital status, occupation, industry, race, sex, citizenship, tax filing, household

Special

Instance weights (col 24): each record represents thousands of real Americans — used for population statistics, excluded from model features



Each record represents thousands of real people

The 6.2% Problem: Why Accuracy Is Misleading

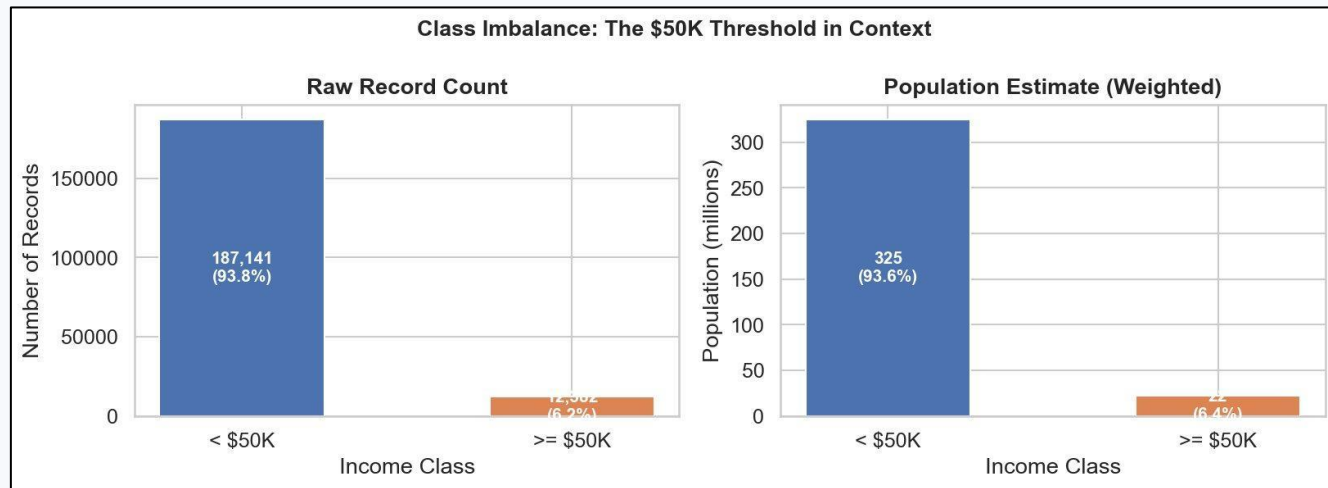
93.8%

Income < \$50K

6.2%

Income ≥ \$50K (~22M Americans)

Use ROC-AUC & PR-AUC, not accuracy



KEY INSIGHT A model that always predicts '<\$50K' scores 93.8% accuracy — and is completely useless. Class imbalance drives every modelling decision downstream.

Data Quality: Migration Columns Require Special Treatment



Migration cols (50%)

Structural missingness — these records belong to non-movers for whom migration data is undefined, not randomly absent. Imputation would be inappropriate.

Country of birth (1–3%)

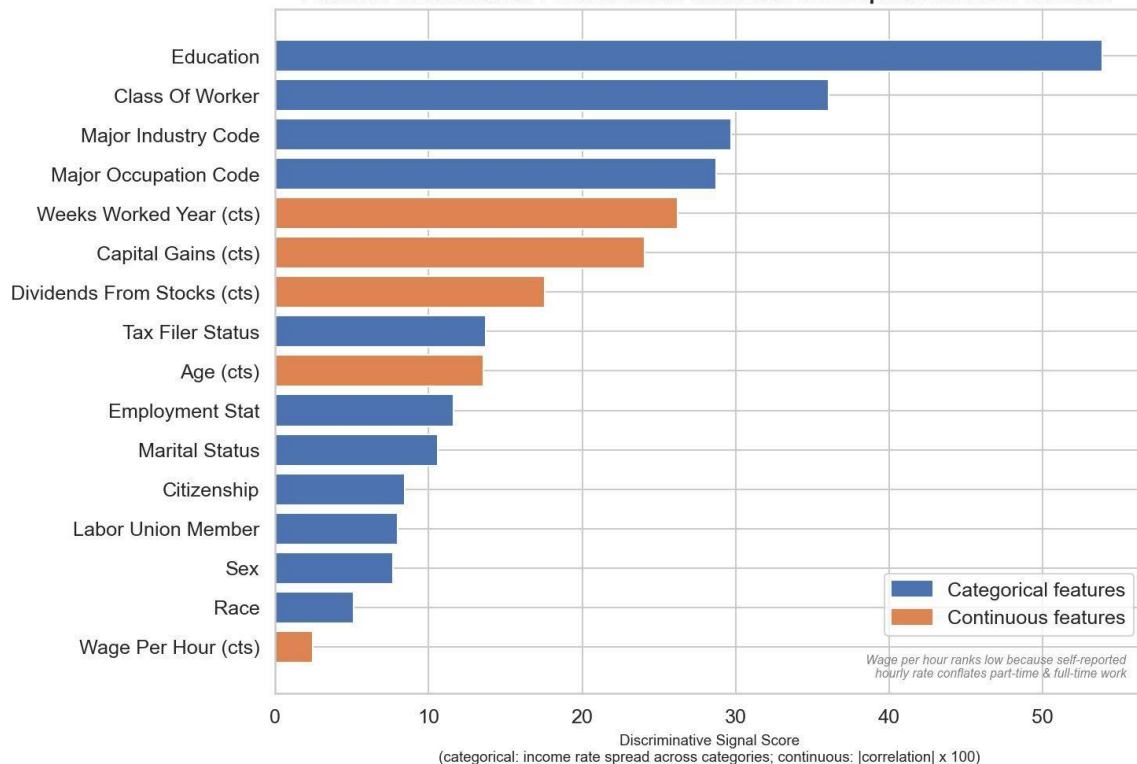
Minor missingness, likely MCAR. Standard imputation (mode for categoricals) is appropriate.

Strategy

Migration columns: add binary 'is_mover' indicator + fill 'Not_in_universe'. Country cols: mode imputation. No imputation for model features that are fully observed.

What Predicts Income? A Roadmap of the Top Signals

Feature Discriminative Power: Which Variables Best Separate Income Classes?



#1 Education

Largest single discriminator. The bachelor's degree is the structural inflection point.

#2 Class of Worker

Government vs self-employed vs private sector differ substantially.

#3–4 Industry & Occupation

Sector matters as much as seniority.

Capital Income

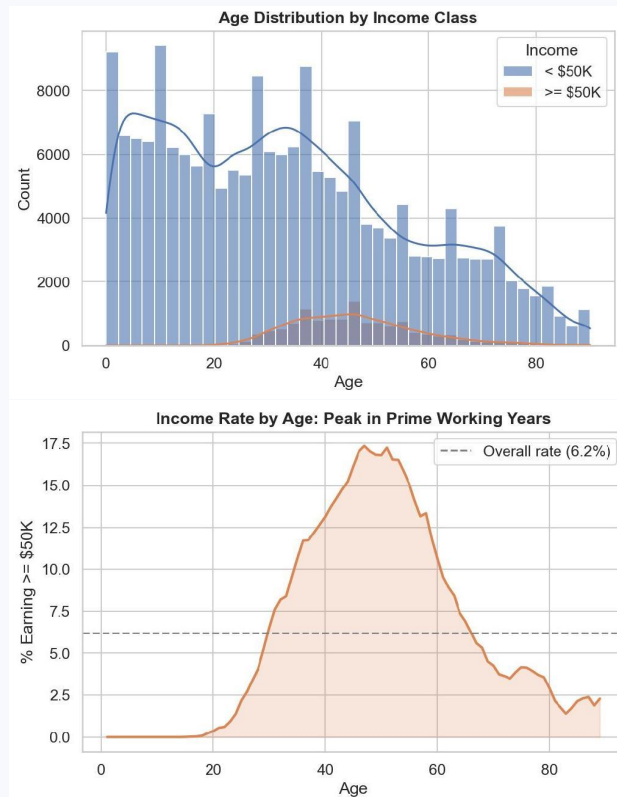
Despite covering only 3.7% of population, capital gains/dividends signal high earning with 6x lift.

Low: Wage/hr

Self-reported hourly rate conflates part-time and full-time — annual weeks worked is far more informative.

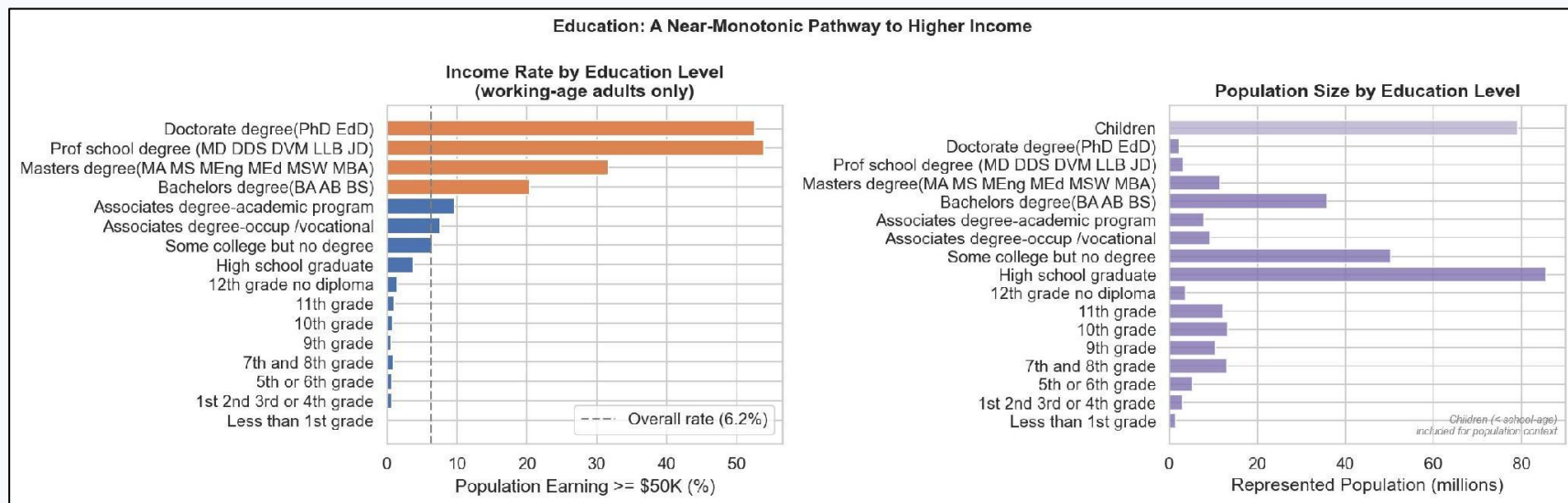
Portrait of a High Earner: Who Are the 22 Million?

Attribute	High Earner ($\geq \$50K$)	Typical American
Age	45–54 years (peak income decade)	35–44 years
Education	Bachelor's degree or higher	High school diploma
Marital Status	Married, civilian spouse present	Never married / other
Employment	Full-time, full year (52 weeks)	Variable
Capital Income	Present — gains or dividends	Absent (96% of population)
Weeks Worked	~52 weeks (full year)	~40 weeks



KEY INSIGHT High earning is not one trait — it's a compound of education, age, work attachment, and asset ownership that reinforce each other. Income rate peaks in the 45–54 age decade, then declines toward retirement

Finding 1 — The Education-Income Ladder



% Population earning >= \$50k

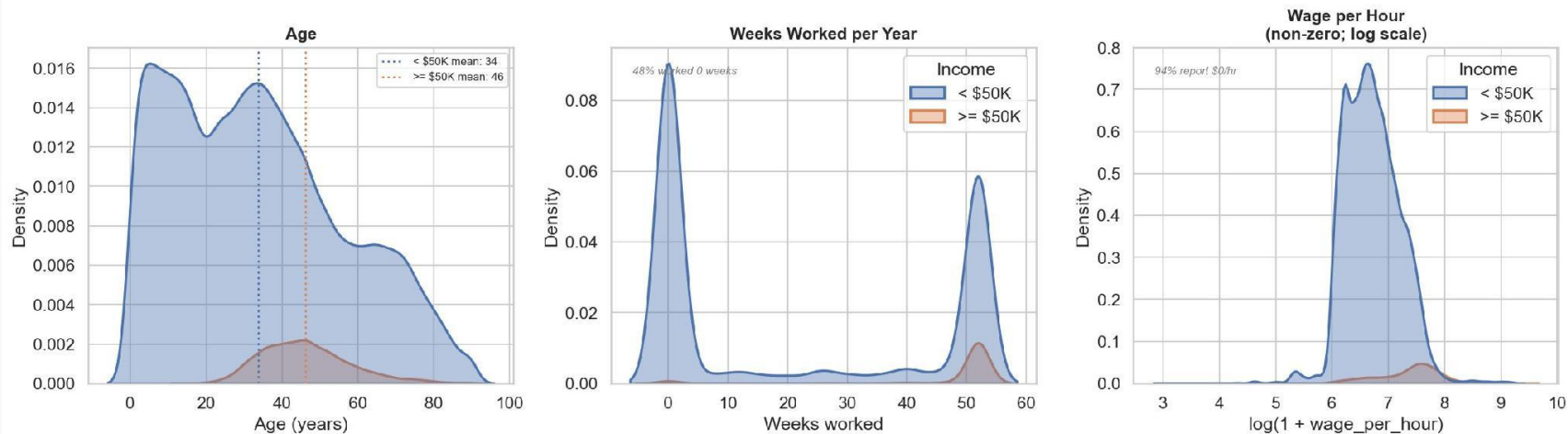
3–4% · High school grad income rate (near population average of 6.2%)

~20% · Bachelor's degree — the major structural inflection point

~52% · Doctoral/professional degree — double the bachelor's rate

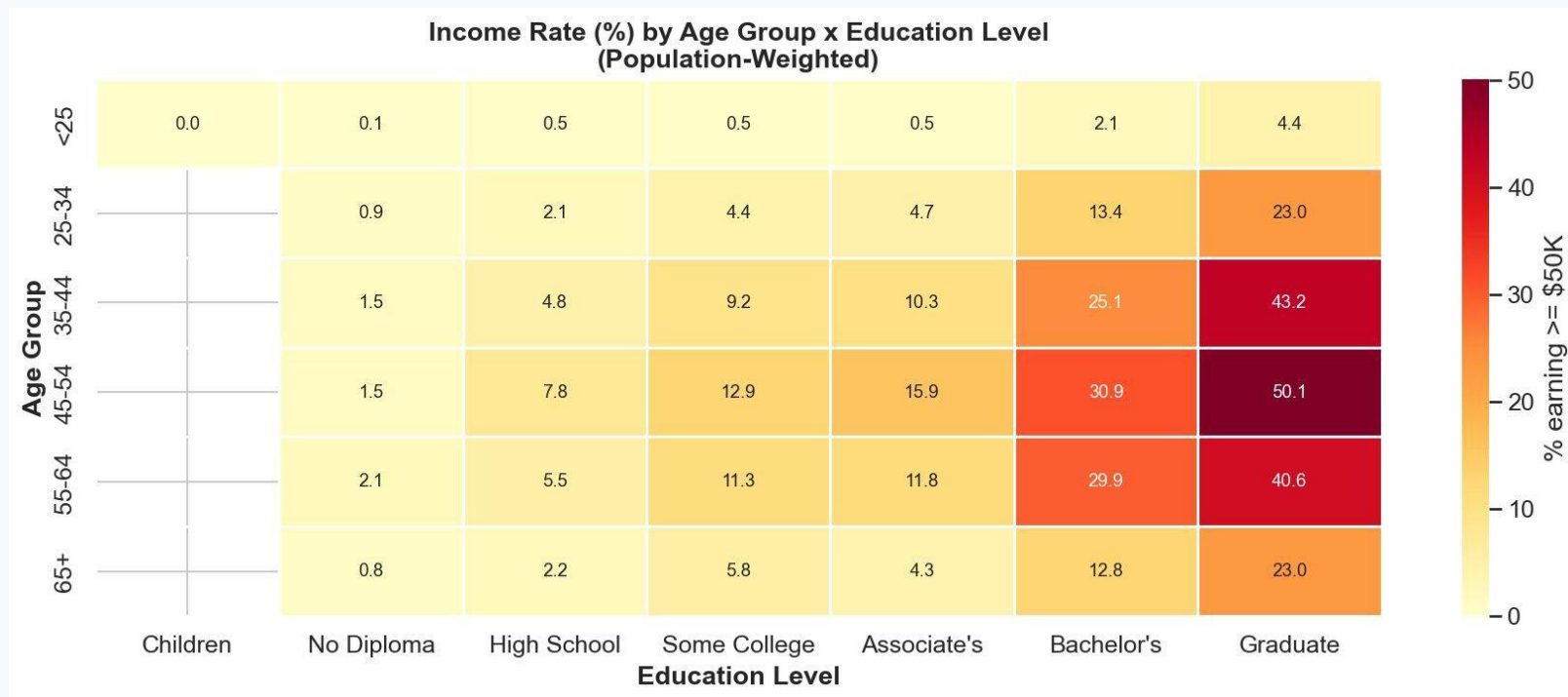
Finding 2 — Age as a Life cycle Income Predictor

Key Continuous Features: High Earners Are Older, Work More Weeks, and Earn More per Hour



KEY INSIGHT High earners are ~12 years older on average (mean 46 vs 34). They also work significantly more weeks per year — 48 vs 22. Age and work intensity compound each other.

Finding 2b — Age × Education: The Compound Effect

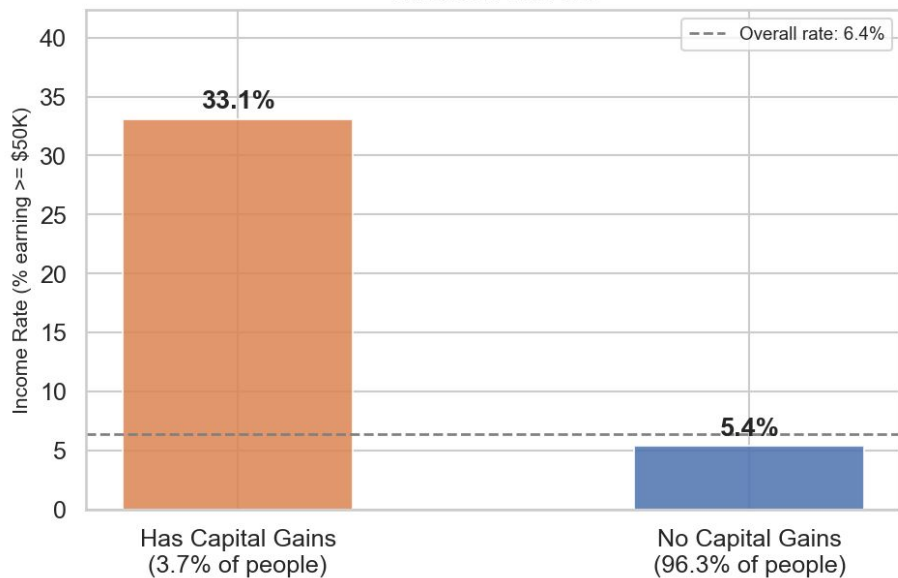


KEY INSIGHT A 45–54 year old with a graduate degree has a 50.1% income rate — 8× the population average. Age and education interact multiplicatively, not additively.

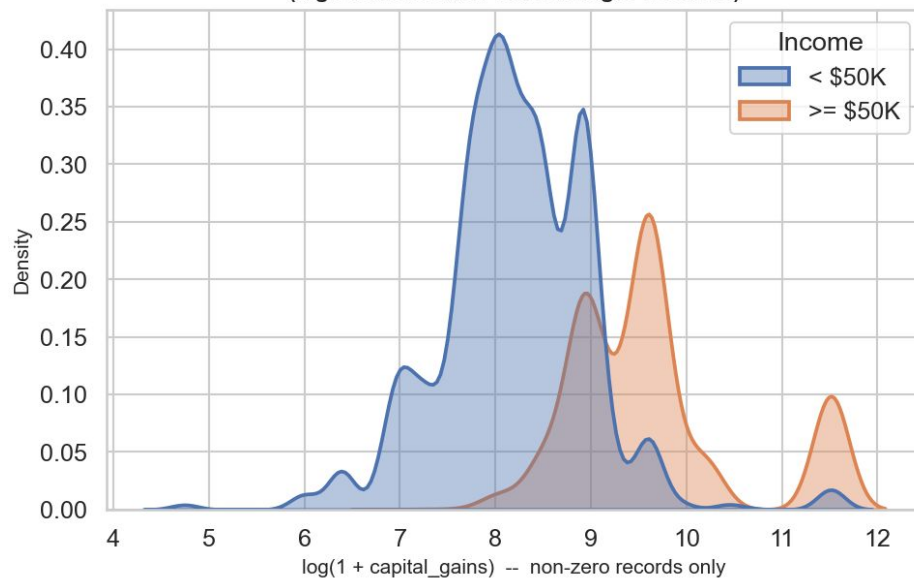
Finding 3 — Capital Income: Sparse but Extraordinarily Powerful

Capital Income: Sparse (3.7% of People) but the Sharpest Discriminator in the Dataset

Capital Gains Presence:
6x Income Rate Lift



Distribution of Non-Zero Capital Gains
(high earners skew toward larger amounts)



3.7%

Only 3.7% of records have capital gains

33% vs 5%

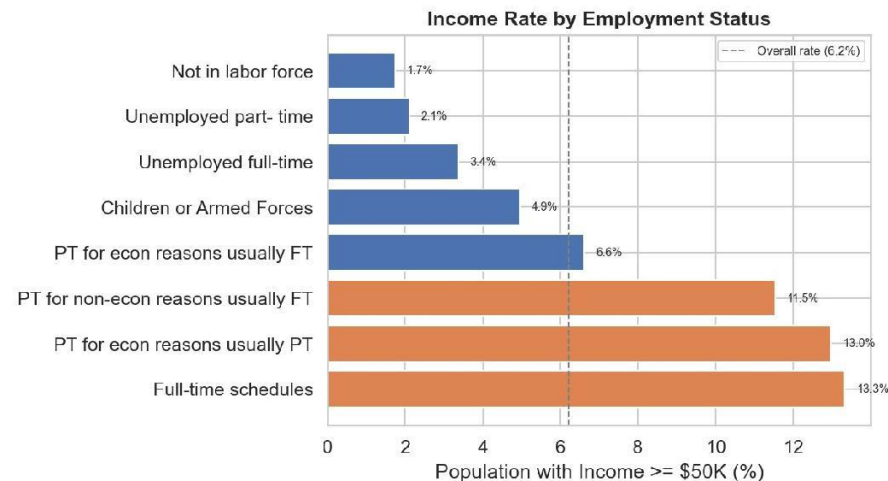
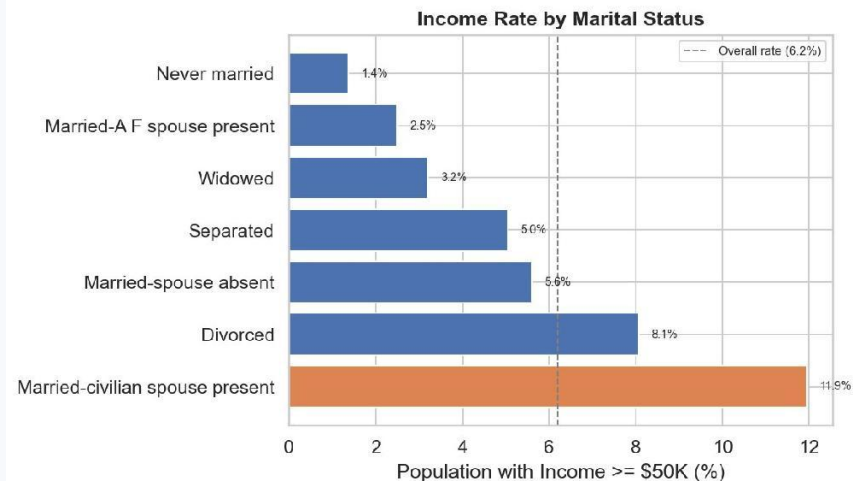
High Income rate for those with vs. without capital gains — 6.6× lift

Binary signal

Presence matters more than amount — a binary flag outperforms the raw value

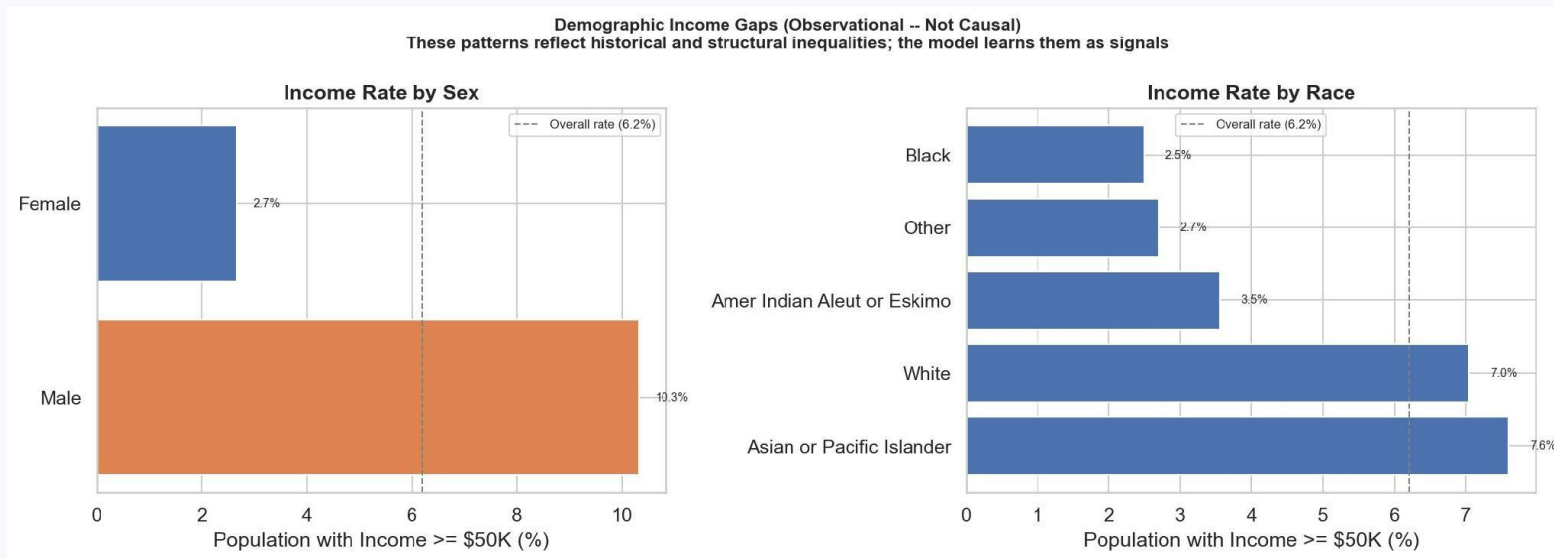
Finding 4 — Household Structure and Work Status as Income Signals

Household Structure and Work Status as Income Signals



KEY INSIGHT Married-civilian-spouse-present households have 11.9% high income rate — nearly double divorced, and 8× never-married. Full-time workers at 13.3% vs not-in-labor-force at 1.7%. Household structure proxies for career stage and economic stability.

Finding 5 — Demographic Income Gaps (Observational, Not Causal)



⚠ Important caveat: These patterns reflect historical structural inequalities in 1994–95. The model learns them as statistical signals. In a real deployment, formal fairness audits (demographic parity, equal opportunity) and possible post-hoc recalibration would be required.

Finding 5 — Demographic Income Gaps (Observational, Not Causal)



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Three Models: LightGBM Wins on What Matters

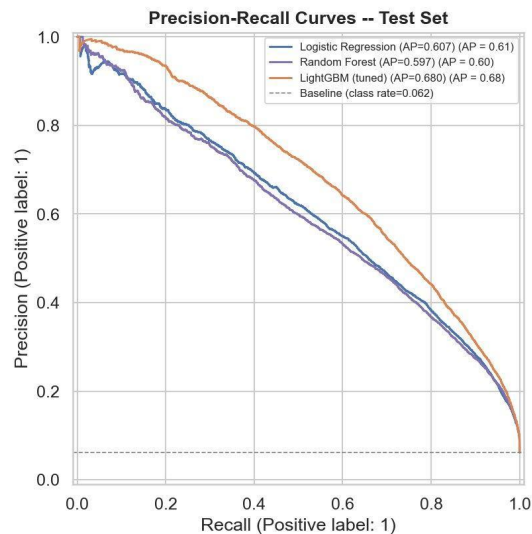
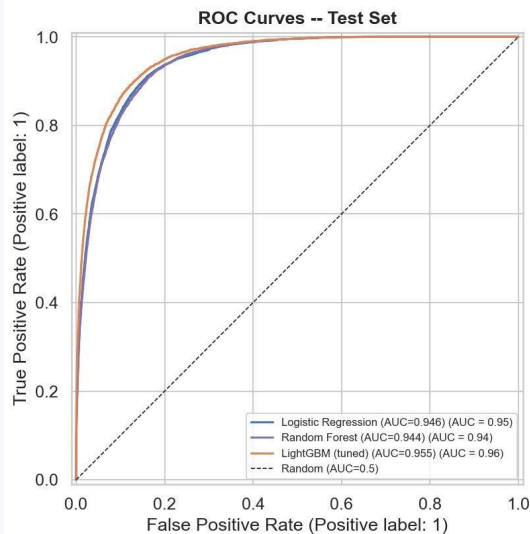
Model	ROC-AUC	PR-AUC	Notes
Logistic Regression	0.946	0.607	Linear baseline, fast to train, strong interpretability
Random Forest	0.944	0.596	Ensemble baseline, captures non-linearity
LightGBM (tuned) ★	0.955	0.677	Gradient boosting, Optuna-tuned — selected model

Why PR-AUC is the decisive metric

- All three models are close on ROC-AUC (0.944–0.955) — ROC is less informative on imbalanced data
- LightGBM leads by +7 points on PR-AUC (0.677 vs 0.596–0.607)
- PR-AUC directly measures the model's ability to rank the 6.2% minority class
- 5-fold stratified cross-validation used throughout to prevent data leakage

LightGBM Results: ROC-AUC 0.955 · PR-AUC 0.677

Model Comparison: LightGBM Achieves Best Performance on Both Metrics



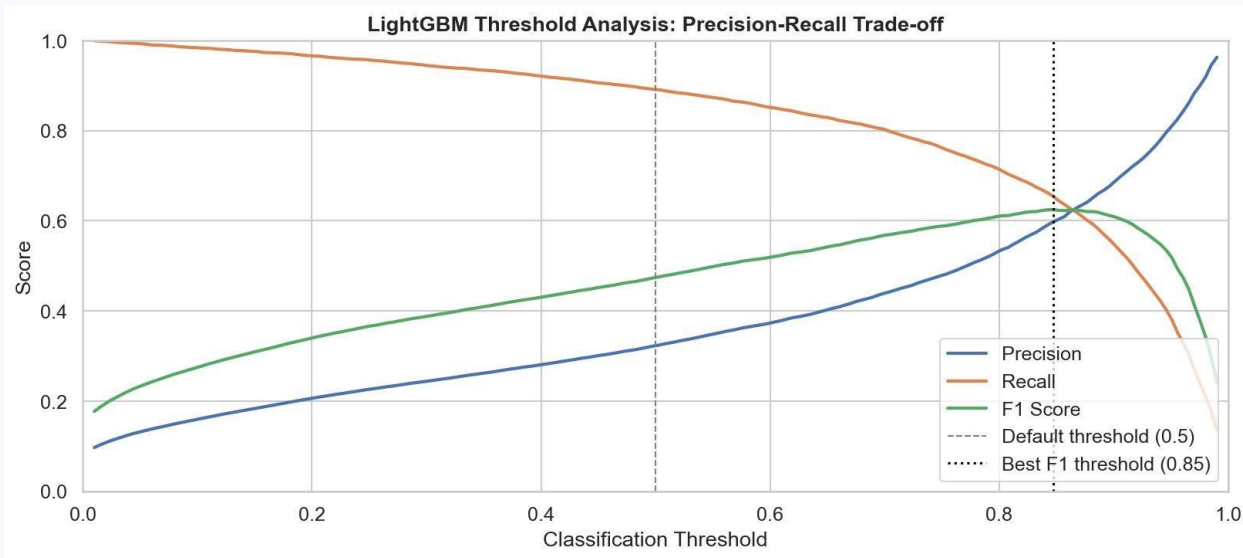
0.955
ROC-AUC

0.677
PR-AUC

0.857
F1-optimal threshold

89.6%
Recall at threshold 0.857

Choosing the Operating Threshold: A Business Decision



KEY INSIGHT Threshold choice is not purely technical — it reflects the cost of false negatives vs false positives for the specific use case.

Lower threshold → Higher Recall

Flag more qualifying households.
More false positives, fewer missed cases.

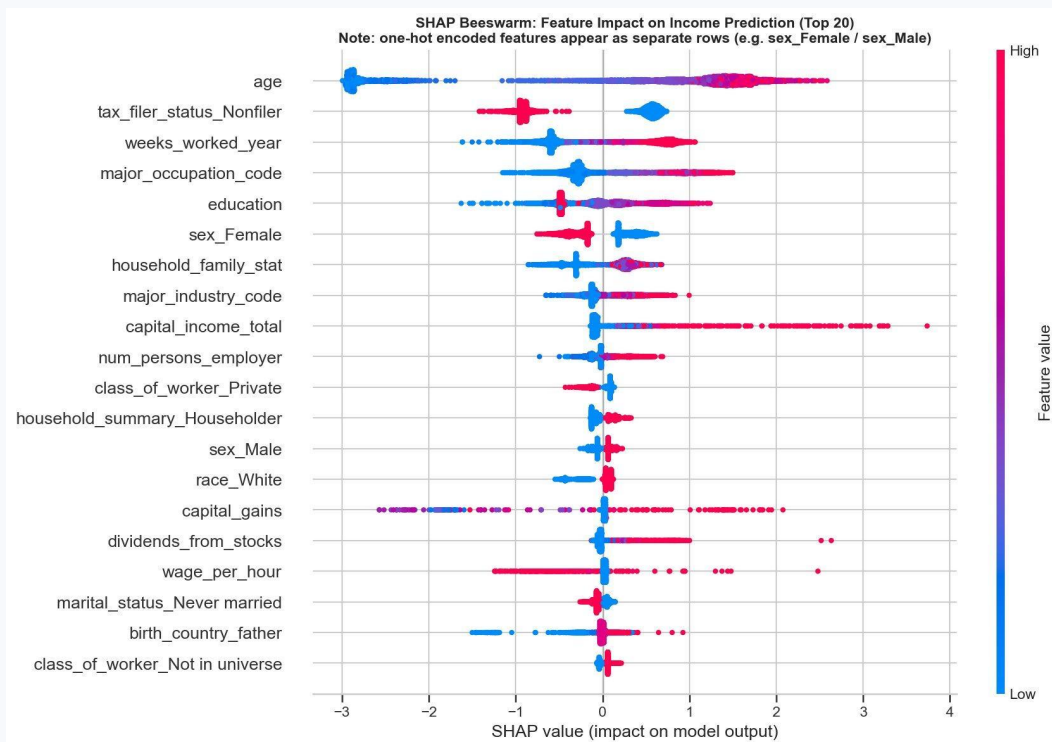
Preferred for Census Bureau resource allocation — missing a qualifying household is costlier than a false positive that triggers manual review.

F1-optimal (0.857)

Best balance of precision and recall.

At 0.857: Recall 89.6%, Precision 31.7%, F1 0.468. High threshold reflects calibration — the model outputs low probabilities because positives are rare.

What the Model Learned: SHAP Feature Importance



EDA ↔ SHAP alignment

Education, age, occupation dominate both — the model learned real patterns, not noise.

age (top feature)

Red = older → strong positive SHAP. Confirms the lifecycle S-curve from EDA.

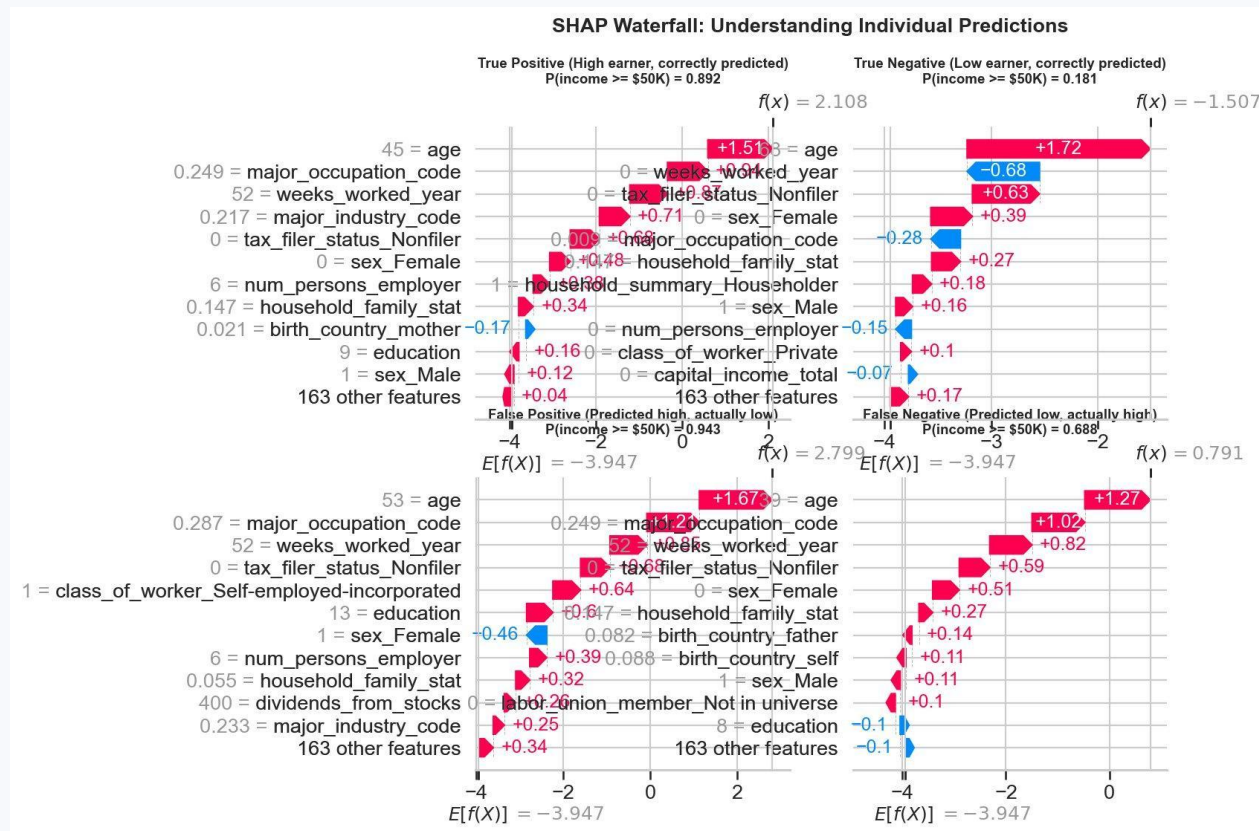
Two surprises

Tax filer status and weeks_worked capture labor force attachment interactions invisible in univariate analysis.

Sex encoding

sex_Female and sex_Male appear as separate rows (one-hot artifact) — they represent one underlying signal.

Inside a Prediction: How the Model Explains Its Choices



Red bars push toward $\geq \$50K$. **Blue bars** push toward $< \$50K$. Starting point: 6.2% base rate. **False negatives** = early-career high earners whose income signal isn't yet visible in these features.

Key Takeaways and Limitations

✓ KEY FINDINGS

1. Education is the clearest structural predictor

The bachelor's degree jump is the single largest income threshold. Graduate degrees roughly double the odds again.

2. Capital income is the sharpest discriminator

Binary flag with 6× lift, despite covering only 3.7% of records. Presence matters more than amount.

3. LightGBM best captures compound effects

Age × work intensity × education interactions are invisible in univariate analysis but captured well by gradient boosting (ROC-AUC 0.955, PR-AUC 0.677).

⚠ HONEST LIMITATIONS

1. Data is 30 years old

\$50K in 1994 ≈ \$100K in 2024 dollars. Findings are directionally valid but magnitudes and some demographic patterns have shifted considerably.

2. All findings are observational associations

The model predicts income but makes no causal claims. Correlation ≠ causation, especially for demographic features.

3. Random train/test split, not temporal

A year-based split (1994 train / 1995 test) would better simulate real deployment conditions and is worth exploring as a robustness check.

Questions & Discussion

Thank you for your time.
